30 Interview Questions and Answers

Multiple PDF RAG Assistant | Getamesay Zegeye

1. What problem does this project solve?

It answers questions across uploaded PDFs while exposing retrieved passages and their file/page metadata.

2. What does RAG mean here?

Retrieval selects document passages, then a language model generates an answer using that context.

3. How does multi-PDF upload work?

Streamlit accepts multiple PDF files and extract_chunks iterates over each file and its pages.

4. Why preserve page metadata?

It makes source evidence traceable and supports file/page citations in answers.

5. How are chunks created?

Each page uses 1,500-character windows with a 1,300-character stride, giving 200-character overlap.

6. Why use overlap?

It retains nearby content at boundaries, although it may add redundancy and still split a sentence.

7. What happens to empty pages?

They are skipped; a collection with no readable text raises a clear processing error.

8. Does it support scanned PDFs?

Not directly. pypdf extracts existing text; scanned images require an OCR stage first.

9. Which embedding model is configured?

gemini-embedding-001 through the Google Gen AI SDK. Both document and question embeddings call the API.

10. Why use different embedding task types?

RETRIEVAL_DOCUMENT identifies indexed text and RETRIEVAL_QUERY identifies a search question for the embedding service.

11. Why batch embeddings?

Groups of up to 20 chunks support progress reporting and bound the number of inputs in each application-level call.

12. Why convert vectors to float32?

FAISS expects suitable numeric arrays; float32 is the standard representation used by this implementation.

13. Why normalize vectors?

Unit-length vectors let inner-product ranking correspond to cosine similarity.

14. Which FAISS index is used?

IndexFlatIP, an exact inner-product index; it does not use approximate nearest-neighbor search.

15. Is the FAISS index persistent?

No. It is stored in Streamlit session state and must be rebuilt after the session is lost.

16. How many chunks are retrieved?

Up to six, or fewer if the index contains fewer than six vectors.

17. Does retrieval guarantee both PDFs are represented?

No. A global top-six search may return passages from only one file.

18. How could you improve cross-document coverage?

Measure coverage first, then compare per-document candidates, decomposed queries or hybrid retrieval under a fixed context budget.

19. Which language model generates answers?

The configured alias is gemini-flash-lite-latest; aliases can change over time, so record the resolved model for reproducibility.

20. What grounding instructions are used?

Answer from reference passages, address requested parts, cite filename/page, allow supported comparisons and identify specific missing facts.

21. Are citations automatically verified?

No. Citations are generated by the model and should be checked against displayed passages.

22. How do you diagnose a false missing-information answer?

Inspect the actual retrieved context, check extraction and source coverage, then assess whether generation ignored available evidence.

23. How does the app prevent stale-document queries?

It compares the indexed selection with tuples of filenames and SHA-256 file-content hashes before enabling the chat input.

24. What happens if processing fails midway?

The old session index remains because replacement occurs only after build_index succeeds. Completed API calls may still consume quota.

25. Does the model remember earlier chat messages?

No. Messages are displayed, but only the current question and retrieved passages enter the generation request.

26. What request limits are configured?

A 60,000 ms client timeout, one configured attempt and a 1,000-token generation output limit.

27. How should a 429 embedding error be handled?

Inspect the quota metric and retry guidance, avoid repeated reprocessing and consider persistence or alternative embeddings after measuring needs.

28. What data leaves the computer?

Chunk text goes to Google for embeddings; questions go for query embedding and generation; retrieved context also goes for generation.

29. How would you evaluate the app?

Use frozen questions with direct, cross-document and absent-answer cases; score retrieval coverage, answer correctness, completeness, citations and abstention separately.

30. What would you change before shared deployment?

Add data isolation and access controls, persistent index management, bounded inputs/context, quota-aware handling and exact-source integration testing.